

Disanda Epaladeniya

Final Year Undergraduate Student

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📁 Portfolio

PROFILE

I am a dynamic final year undergraduate student specializing in Electrical and Information Engineering, eager to apply my skills and contribute to innovative projects while gaining practical experience. I have a strong interest in Digital Electronics, FPGA Design, and Embedded Systems Design. With a solid foundation in problem-solving, a collaborative mindset, and a passion for continuous learning, I strive to be a reliable team player and a well-rounded engineering professional

PROFESSIONAL EXPERIENCE

National Engineering Research & Development Centre (NERDC)  08/2025 – 10/2025
Trainee Electronics Engineer

- Designed a driver circuit for a buck converter as part of a Solar Battery Charging System with MPPT.
- Developed and validated circuit schematics and PCB layouts using industry-standard EDA tools.
- Gained hands-on experience with STM32 micro-controllers, including peripheral configuration.
- Represented at Industry Expo 2026

Tools Used: STM32 Cube IDE | Altium Designer | KiCad

Sri Lanka Telecom PLC, *Trainee Telecommunication Engineer* 06/2024 – 08/2024

- Gained hands-on experience in telecommunication and networking
- Had exposure to technologies such as fiber-optic communication, ADSL, and voice communication systems.
- Engaged in a circuit design project, and had the opportunity to have direct involvement in customer and employee interactions.

EDUCATION

University of Ruhuna  05/2022 – 08/2026 | Galle, Sri Lanka
B.Sc (Hons.) in Electrical & Information Engineering

OGPA : 3.61/4.00

Special Modules Followed:

Introduction to Electronic Engineering | Analog Electronics | Digital Electronics | Electronic Circuit Design | Computer Architecture | Hardware Description Language | Embedded Systems Design | Operating Systems Programming | Control Systems Design | High Performance Computing | Digital Signal Processing | Machine Learning | Intelligent Systems Design

SKILLS

C/ C++ | Python | Verilog | System Verilog | Vivado | Matlab | Altium Designer

COURSES

Learning FPGA Development/ Learning Verilog for FPGA Development, *Linkedin Learning*

- Field Programmable Gate Arrays | Verilog

PROJECTS

16 Bit MIPS Processor, *A Verilog Project*

Implemented a 16-bit MIPS processor that supports arithmetic operations ADD, SUBTRACT, MULTIPLY, DIVISION instructions AND, OR, XOR, XNOR, EQUAL, Greater Than, Less Than logical instructions. Included program counter, instruction memory, data memory following harvard architecture.

- Verilog
- Vivado

Image Processing Driven Line Following Robot with Smart Route Optimization,

A project showcased at ReXtro 2025

The project represents a prototype for an AGV (Automated Guided Vehicle). Has the ability of choosing the shortest path and fast ability of changing its path map by only changing the coordinates of junctions and end points.

- Image Processing through OpenCV
- Turn decision using Vector Product and Graph theory
- Dijkstra's Algorithm to determine shortest path
- Raspberry Pi 5 and Pi Camera module 2 used as main hardware

Multi-UAV Simulation Framework using open source tools, *Final Year Undergraduate Project*

The project is currently on progress which aims to create a framework to simulate multiple UAVs which represents techniques such as clustering, network analysis between UAVs and use of companion computers. This project integrates open source tools such as Gazebo, Ardupilot, ns3 and ROS2.

- Network simulation through ns3.
- 3D physics simulation through Gazebo.
- ROS2 used as middleware.
- Ardupilot used for UAV controlling.

VOLUNTEERING

XbotiX, *Lead (2026)/ Co-Lead (2025) - Partnership Development*

04/2025 – Present

XbotiX is the significant robot competition held in Down South of Sri Lanka organized by the Faculty of Engineering, University of Ruhuna. I am having the opportunity to develop partnerships and communicate with different stakeholders together with team handling to make the event a grand success.

AIESEC in Sri Lanka ☺ , *Local Committee Vice President (2024)/ Analyst (2023)/ Manager (2023)/ Team Leader (2023)/ Team Member (2022)*

2022 – 2025

Led the Incoming Global Talent & Teacher function at AIESEC in University of Ruhuna, where I developed leadership and strategic planning skills by organizing international exchange programs.

REFERENCES

Dr. Thilina Weerasinghe ☺ , *Senior Lecturer,*

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Dr. Subodha Gunawardena ☺ , *Senior Lecturer,*

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